

Professional Power BI Project Documentation

Completed by sridhar.P

Industry: Oil & Gas - Upstream Operations

Documentation is based on the completed dashboard material supplied for this project. Exact source-system names and physical refresh architecture are not specified in the supplied material.

1. Project Title

Shell Upstream Performance & Emissions Command Center

A professional Power BI analytics solution designed to bring upstream production, operations, financial performance, emissions, decarbonization, and scenario analysis into one interactive management command center.

2. Project Overview

The Shell Upstream Performance & Emissions Command Center is an interactive Microsoft Power BI dashboard created to provide a consolidated management view of upstream production, operations, financial performance, emissions, decarbonization performance, and strategic scenarios.

Executive Overview - management-level summary of production, finance, and emissions.

Production & Operations - production, assets, downtime, maintenance, and operational trends.

Financial Performance - revenue, cost, margin, regional and segment revenue, and Brent-price context.

Emissions & Decarbonization - Scope 1, Scope 2, Scope 3, regional emissions, targets, and emissions composition.

Scenario Analysis - comparison of oil-price, growth, revenue, emissions, and renewable-share assumptions.

The completed dashboard contains five report pages and is intended for management, operations, finance, sustainability, and planning users. ■filecite■turn0file0■L10-L20■

3. Industry - Problem Statement

Industry: Oil & Gas - Upstream

The upstream oil and gas business involves complex operational, financial, asset, commodity-price, and environmental information. Production performance, asset availability, downtime, maintenance, revenue, cost, margin, emissions, and future scenarios can be difficult to evaluate together when they are reviewed separately.

The project addresses this problem by consolidating major performance measures into one interactive command center. The dashboard helps users understand production by time and region, asset contribution and downtime, maintenance cost, revenue and margin, Brent-price context, emissions scopes and regions, target performance, and scenario outcomes. ■filecite■turn0file0■L65-L81■

4. Tools & Technologies

The supplied documentation confirms Power BI as the primary platform and describes the analytical flow from source/business data through preparation, the Power BI model, calculated measures, visuals, filters, navigation, and management insights. ■filecite■turn0file0■L115-L127■

5. Dataset Introduction and Source

The project dataset represents upstream business information required to analyze production, operations, financial performance, emissions, and scenarios. The dashboard uses dimensions and measures covering time, region, segment, asset, maintenance type, emissions scope, targets, and scenario assumptions.

The supplied project documentation does not specify exact external source-system names, database server, database name, file locations, or refresh architecture. Those details are therefore intentionally not invented.

The dashboard scope and data areas are documented in the supplied project material.

■filecite■turn0file0■L95-L114■

6. MYSQL - Database and Tables

MySQL is used as the database layer in the project workflow. Its purpose is to organize the upstream information into structured data that can be transformed and consumed by Power BI.

The supplied documentation does not contain the exact physical MySQL database name or table names. To maintain accuracy, the following is a logical table organization rather than an invented physical schema.

If the actual MySQL table names are available in the SQL project, they can be inserted into this section without changing the document structure.

7. Connecting MYSQL with Power BI

The MySQL-to-Power BI connection forms the bridge between structured database information and the Power BI analytical model. The workflow is to connect Power BI to MySQL, select the required data objects, and bring them into Power Query for ETL.

Open Power BI Desktop and choose the MySQL database connector.

Enter the MySQL server and database details.

Authenticate with the required database credentials.

Select the required production, operations, financial, emissions, target, scenario, and date objects.

Choose the appropriate connection/import approach for the deployment.

Open Power Query and begin ETL/data preparation.

Apply transformations and load the prepared tables into the Power BI model.

Create relationships, measures, and visuals.

The overall source-to-model workflow is confirmed by the supplied documentation, while exact server and connection details are not specified. ■filecite■turn0file0■L115-L127■

8. Loading Data to ETL

Power Query acts as the ETL preparation layer between the source data and the final Power BI model.

Extract - connect to MySQL and select the required data.

Transform - clean columns, standardize values, set correct data types, and reshape data where necessary.

Validate - check blanks, duplicates, invalid values, inconsistent categories, and unexpected totals.

Load - load prepared tables into the Power BI data model.

Model - establish relationships and analytical structures required for reporting.

This follows the project's documented source → preparation → model → measures → visuals → filters/navigation → decision-support flow. ■filecite■turn0file0■L115-L125■

9. Data Cleaning

Data cleaning ensures that KPI cards and visuals use consistent and reliable information.

Remove invalid duplicate records where duplication is not a valid business event.

Handle missing or blank values in critical fields.

Assign correct data types to dates, numeric measures, percentages, and categories.

Standardize region, segment, asset, maintenance type, scope, and scenario labels.

Validate negative or zero values according to business rules.

Check month and year fields used by trend charts and filters.

Standardize Scope 1, Scope 2, and Scope 3 labels.

Validate year-to-target mappings.

Check scenario assumptions for consistent representation.

Reconcile important totals after cleaning.

The completed project documentation also recommends validation of KPI values, filters, navigation, labels, table width, colors, scope differentiation, related totals, tooltips, and display settings.

■filecite■turn0file0■L369-L381■

10. Basic DAX Measures

DAX measures convert prepared data into reusable business calculations. The exact DAX expressions are not included in the supplied documentation, so the following are professional example measures aligned with the documented KPIs.

The project documentation confirms calculated measures/KPIs and numerator/denominator logic for percentage KPIs. It also documents a selected-year approach for retrieving the corresponding reduction target. ■filecite■turn0file0■L354-L368■

Note: these are documentation-level examples. Replace table and column names with the actual names in the final Power BI model.

11. Dashboard Design - Pages Overview

The report contains five pages with a persistent left-side navigation panel. Each page uses a consistent hierarchy of header/title, KPI cards, analytical visuals, and navigation.

■filecite■turn0file0■L128-L138■

12. Dashboard Pages / Visuals

12.1 Page 1 - Executive Overview

Purpose: provide a management-level snapshot of overall performance across production, finance, and emissions.

The Executive Overview balances operational scale, financial results, and environmental performance.

■filecite■turn0file0■L139-L161■

12.2 Page 2 - Production & Operations

Purpose: analyze production, asset contribution, downtime, maintenance categories, and maintenance cost.

The documented operational insight is that corrective maintenance is a major downtime category and the highest-cost maintenance category, suggesting an opportunity to investigate reliability, root causes, and preventive-maintenance effectiveness. ■filecite■turn0file0■L162-L187■

12.3 Page 3 - Financial Performance

Purpose: analyze revenue, cost, margin, regional/segment performance, and Brent-price context.

The supplied documentation cautions that the Revenue vs Brent Price relationship should be treated as association rather than proof of causation. ■filecite■turn0file0■L188-L215■

12.4 Page 4 - Emissions & Decarbonization

Purpose: monitor emissions, emissions composition, regional emissions, targets, and decarbonization priorities.

The key documented insight is that Scope 3 is the dominant displayed emissions category, so decarbonization analysis should consider value-chain and indirect emissions as well as direct operational emissions. ■filecite■turn0file0■L216-L242■

12.5 Page 5 - Scenario Analysis

Purpose: compare alternative scenarios and understand trade-offs between oil price, production growth, revenue, emissions, and renewable share.

PBIX alignment note: two Scenario Analysis visuals retain their existing Power BI labels while using different underlying fields/measures. This document describes the report as-built rather than changing the dashboard.

Documented scenario assumptions include Accelerated Transition (1% production growth, 30% renewable share), Base (3%, 10%), Growth Scenario (8%, 15%), High Oil Price (5%, 12%), and Low Oil Price (-2%, 15%). The documentation records the Scenario Analysis page exactly as configured in the supplied PBIX; no Power BI changes are assumed.

13. Filters & Slicers

Filters and slicers allow users to change the analytical context without creating separate reports.

The supplied PBIX contains two actual slicer visuals: Scenario on the Scenario Analysis page and Year on the Emissions & Decarbonization page. Other dimensions such as region, asset, and segment are used in report visuals and can participate in visual cross-filtering, but they are not configured as standalone slicer visuals in the supplied PBIX.

14. Key Insights

14.1 Operational Insights

Europe is the highest displayed production region at approximately 204M.

Production varies by month, making trend monitoring useful for identifying peaks and low-output periods.

Corrective maintenance is a major displayed downtime category and the highest displayed maintenance-cost category.

High-downtime assets can be prioritized for reliability and root-cause analysis.

14.2 Financial Insights

Displayed total revenue is approximately 109.05bn and displayed total margin is approximately 55.50bn.

Revenue is distributed across business segments and regions.

Brent price provides useful context for interpreting upstream revenue movements.

Revenue and margin should be monitored together because revenue growth does not automatically imply stronger profitability.

14.3 Emissions Insights

Displayed total emissions are approximately 106.0M.

Scope 3 is the largest displayed emissions component at approximately 76.9M.

High-emitting assets can be prioritized for emissions-reduction investigations.

Actual-versus-target analysis supports environmental performance monitoring.

14.4 Strategic Insights

Growth Scenario has the highest displayed production growth at 8%.

Accelerated Transition has the highest displayed renewable share at 30%.

Scenario comparison makes strategic trade-offs visible.

Scenario results should be used for planning and comparison, not treated as guaranteed forecasts.

These insights are directly documented in the supplied project material.

■filecite■turn0file0■L327-L353■

15. Project Workflow

Business requirement identification - define the questions management needs to answer.

Dataset preparation - identify required production, operations, financial, emissions, target, scenario, and date information.

MySQL database layer - organize source data into structured tables.

MySQL to Power BI connection - connect the database to Power BI.

ETL / Power Query - extract, transform, clean, validate, and shape the data.

Data modeling - establish dimensions, relationships, and analytical structure.

DAX development - create reusable KPI and business measures.

Dashboard design - build five pages using KPI cards, charts, tables, filters, and navigation.

Validation - check KPI values, filters, navigation, readability, totals, colors, and interactions.

Business interpretation - identify operational, financial, environmental, and strategic insights.

Final documentation and presentation - explain the complete solution and business value.

This workflow aligns with the documented source → preparation → model → measures → visuals → filters/navigation → decision-support architecture. ■filecite■turn0file0■L115-L125■

16. Industries / Business Applications

Although this project is designed for upstream oil and gas, the analytical architecture can be adapted to other industries requiring integrated operational, financial, environmental, and scenario analytics.

17. Conclusion

The Shell Upstream Performance & Emissions Command Center is a structured Power BI solution that connects upstream operations, financial performance, environmental performance, and strategic scenario analysis in one interactive report.

The project demonstrates an end-to-end BI workflow involving database organization, ETL/data preparation, data cleaning, DAX measures, data modeling, dashboard design, filters and slicers, visual storytelling, validation, and business insight generation.

The completed dashboard enables management to move from high-level KPIs to operational detail, financial analysis, emissions performance, and forward-looking scenarios. It also provides a foundation for future enhancements such as automated refresh, data-quality monitoring, drill-through analysis, deeper Scope 3 analysis, multi-year emissions target tracking, and side-by-side scenario comparison. ■filecite■turn0file0■L394-L404■

Appendix - Project at a Glance